

Day2:

AWS Key Services for Storage Migration

1. AWS Console overview
2. AWS Account types
3. VPC & Subnets
4. Identity and Access Management
5. Elastic Cloud Compute - EC2
6. Simple Storage Service - S3
7. On-Premise and AWS connection options – Hybrid setup
8. Storage Migration
 - AWS Services/Facilities for Data Migration

How to Access AWS?

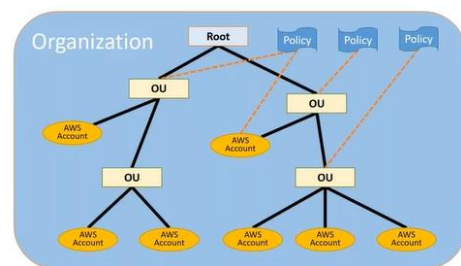
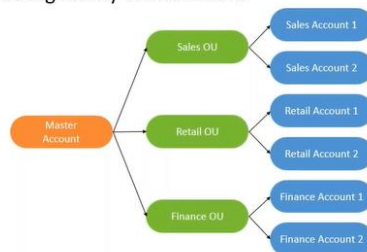
- Create an Individual account with AWS.
- Organization Account

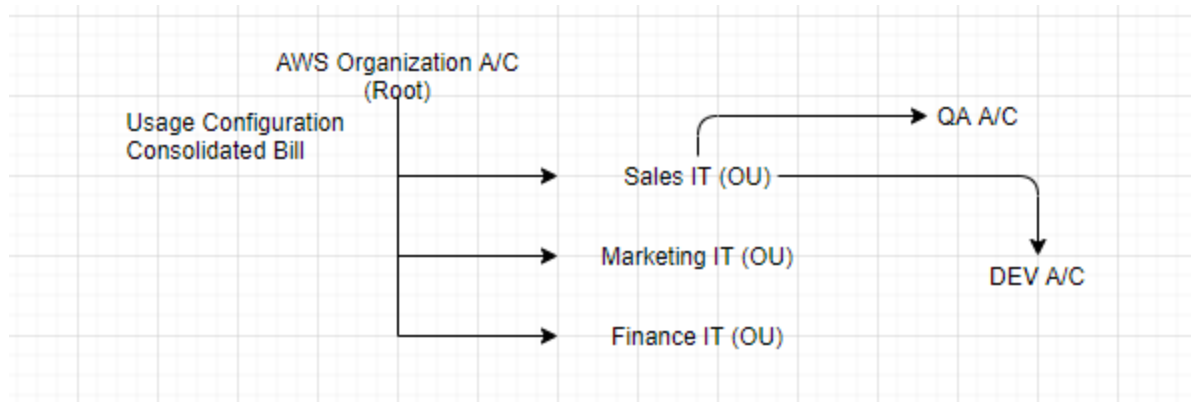
Organization Account:

- AWS Organization-> Services
- AWS Organization-> Policies

- Organization account

- Can have multiple account, one master account
- Hierarchy
 - Multiple accounts under master account
 - Centrally manage all accounts and related policies
 - Volume discounts for EC2 and S3
 - Consolidated billing facility & much more





Architecture Diagram Icons:

VPC

- Subnets
- Internet Gateway
- NACL

Subnets

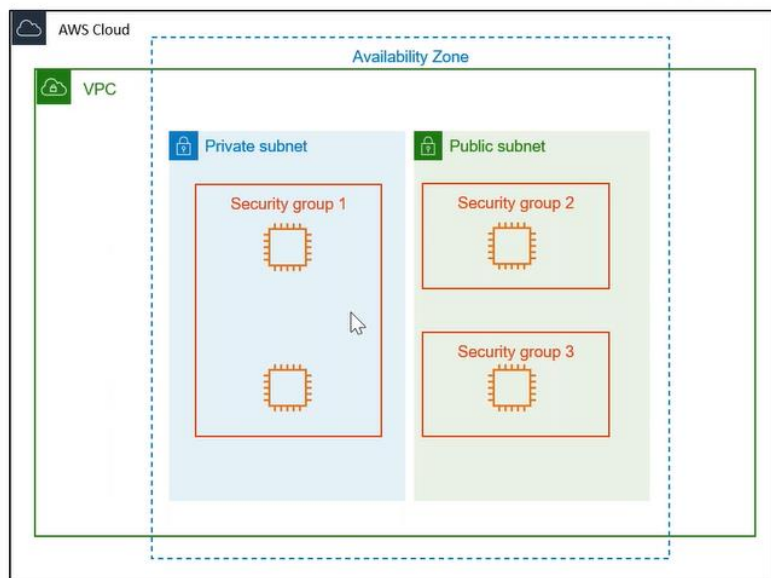
- Private
- Public
- Route table

IAM

- User
- Groups
- Policies
- Role

Launching an EC2

- Instance Types
- Pricing Models
- Vertical Scaling
- Elastic IP
- Security Group



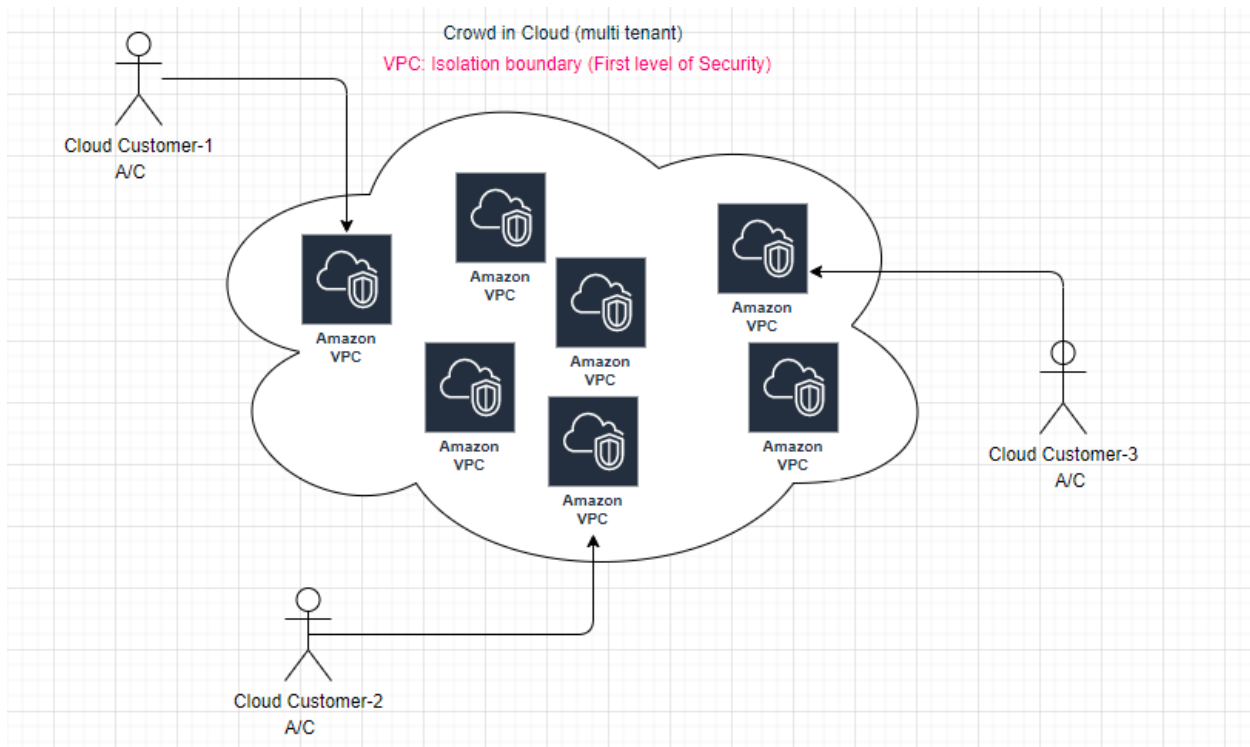
AWS Calculator:

Simple monthly calculator

Setup:

How I Isolate my resources from other's resources?

VPC:



VPC- one default VPC per region

Subnet- subnet corresponding to each AZ.

public subnet: associated with IGW (entry of igw in associated route table), auto assign public ip

private subnet: not associated with IGW (NAT Gateway)

main objective of VPC is to isolate our resources , never use the pre-configured VPC (spans over all AZ with in the region).

IP address range:

With Amazon BYOIP you can bring part, or all your public IPv4 address range from your on-premises deployment to your AWS account. You continue to own the address range, but AWS advertises it on the internet on your behalf.

When using Amazon BYOIP there are several steps involved in bringing your IP address range to AWS, including preparing your range, provisioning, advertising, and allocating Elastic IP addresses.

[Good blog related to BYOIP](#)

Private address range		
Class	start address	finish address
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.31.255.255
C	192.168.0.0	192.168.255.255

Public address range		
Class	start address	finish address
A	0.0.0.0	126.255.255.255
B	128.0.0.0	191.255.255.255
C	192.0.0.0	223.255.255.255
D	224.0.0.0	239.255.255.255
E	240.0.0.0	254.255.255.255

private address don't move between different networks, can assign custom private network during creation time, can't update private network after creation.

step1: create your own VPC (LAN-default 5 vpc per account): private IP range

Create VPC Info

A VPC is an isolated portion of the AWS cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Name tag - *optional*
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block Info

IPv6 CIDR block Info

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

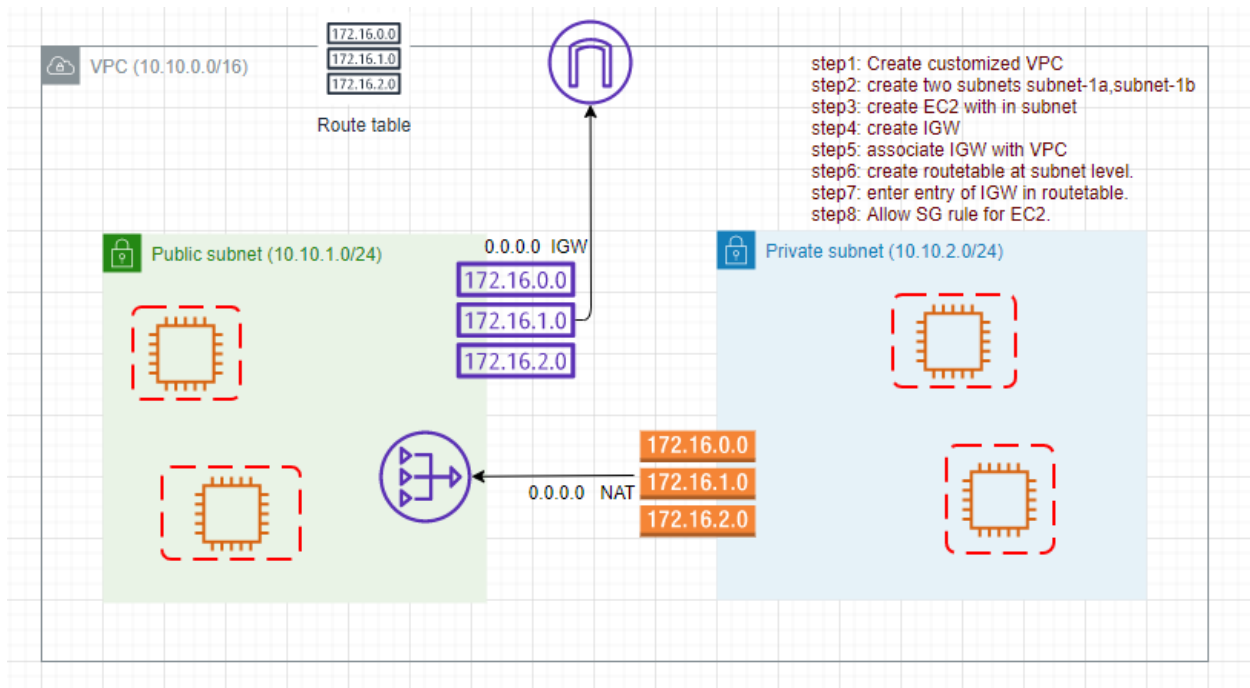
Tenancy Info

step2: By default one subnet per AZ is available, in customized vpc create own two subnets in AZ1 and AZ2, and create EC2 instances with in the subnets.

step3: As of now EC2 is not accessible to the Internet.

step4: Create your own IGW (attach with VPC).

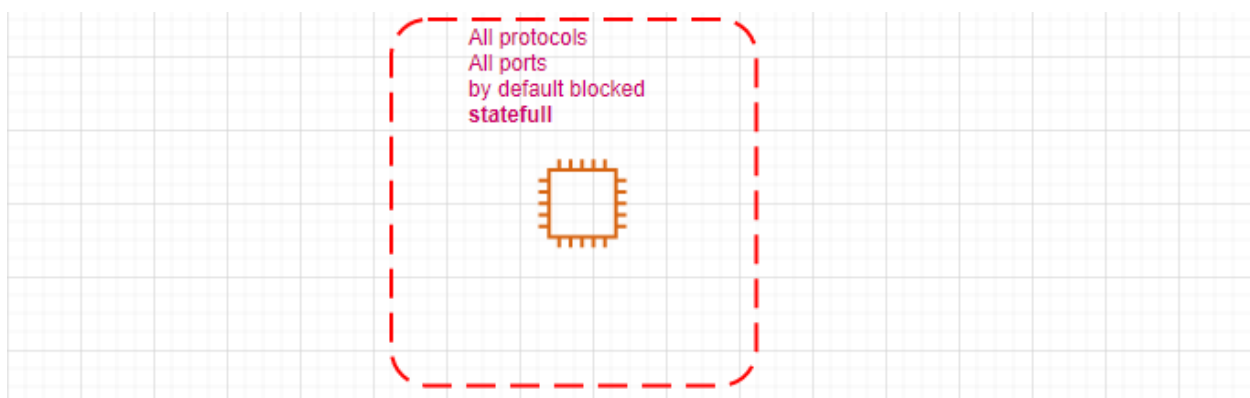
step5: Create route table associated with subnet.



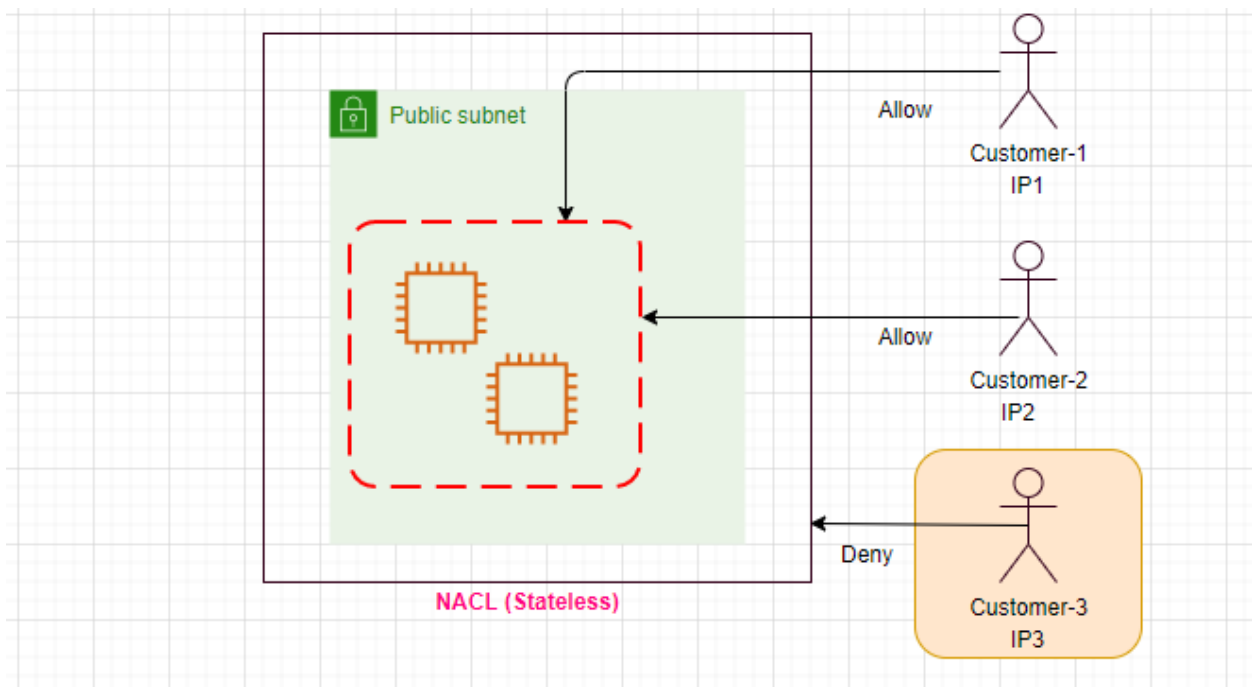
NAT Gateway: EC2 instance inside private subnet access internet through the NAT.

DHCP issues private IP for EC2 .

SG (EC2 level): Deny all traffic by default (Statefull).

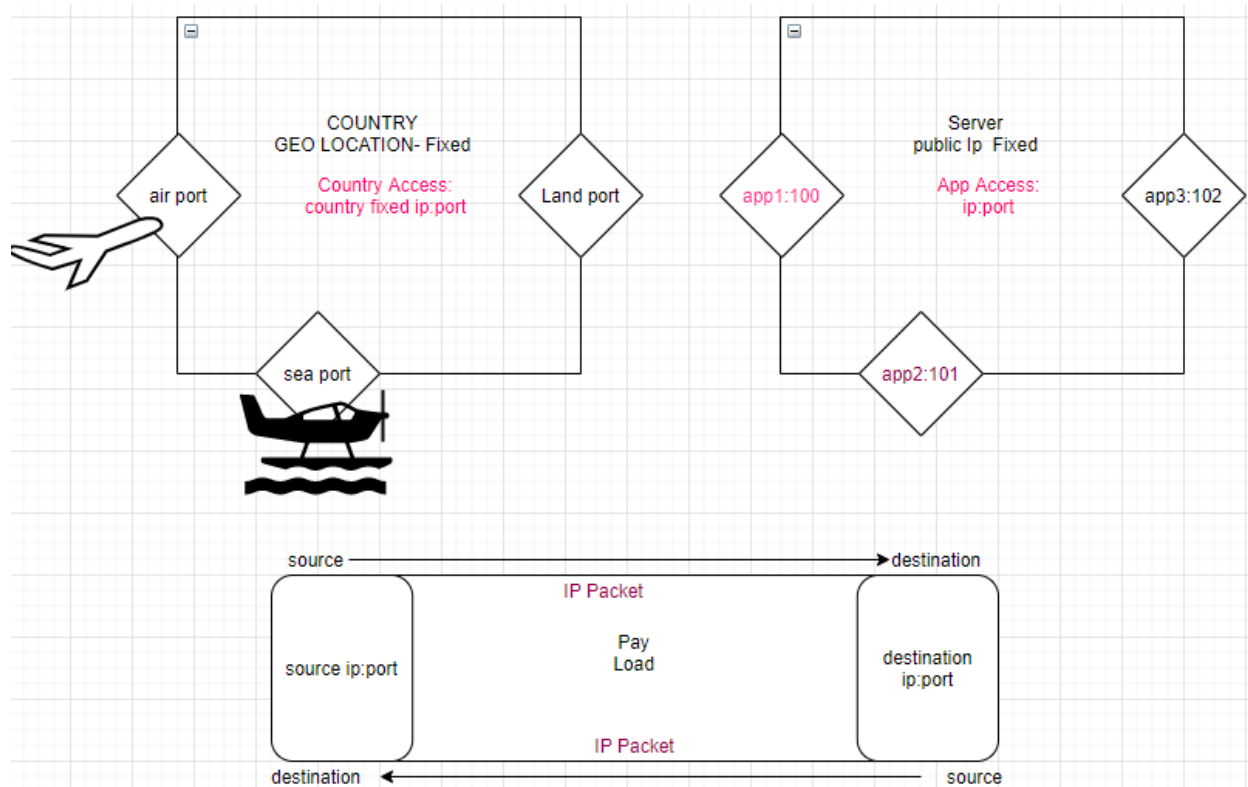


NACL (subnet level): Allows all traffic by default (Stateless). Rules are evaluated sequentially.



ports?

entry point, without port can't run more than one application on the same server.



BYOIP: Bring your own IP, all public IP are paid, by default AWS assigns public ip address from its owned pool of public ip addresses.

Elastic IP: static IP address (associate/release)

status check:

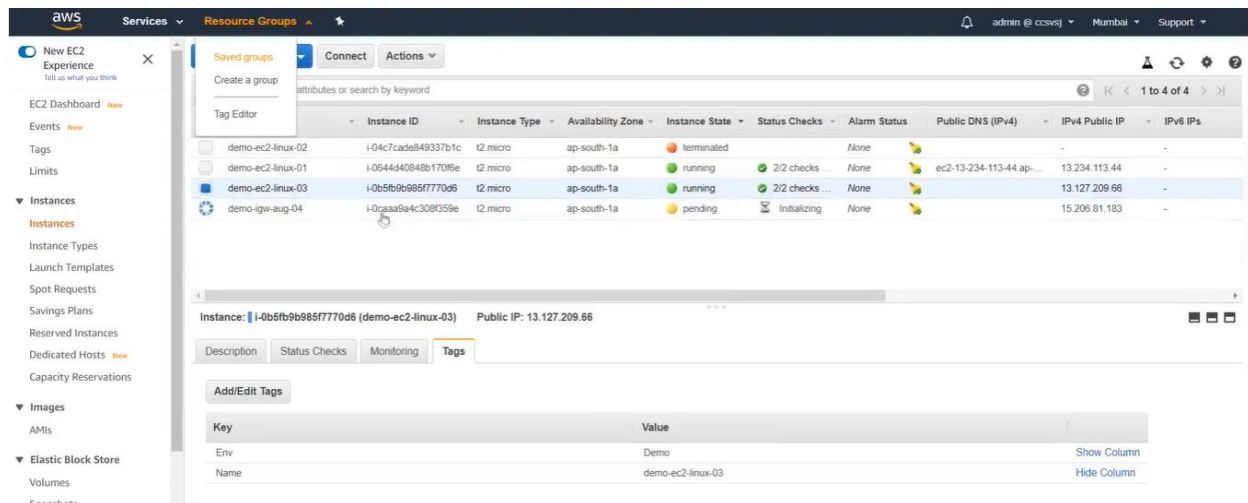
public subnet: - EC2 instances have public as well as private IP addresses.

private subnet: - EC2 instances have only private IP addresses.

CIDR- Classless Inter domain routing

Why tag?

Resource Groups-> Saved Groups



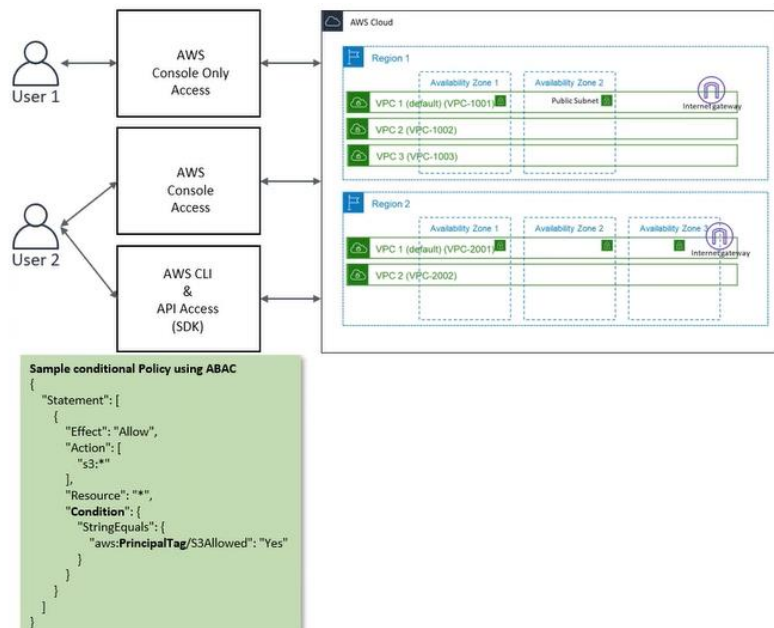
IAM: (who has access to the resources)

IAM (Authorization, Authentication, Users, Groups, Roles, Policy)

ABAC (Attribute based access control): Using tags

Context: Within the Account

- Access Modes
 - Console Only
 - Console + Programmatic
- IAM - [Identity and Access Management](#)
 - Authentication
 - Authorization
 - Users
 - Groups
 - Password Policy
 - Multifactor Authentication
- [Billing Access](#)
- Tags and Usage (ABAC)



step1: IAM-> create a user

Add user

1 2 3 4 5

Set user details

You can add multiple users at once with the same access type and permissions. [Learn more](#)

User name*

+ Add another user

Select AWS access type

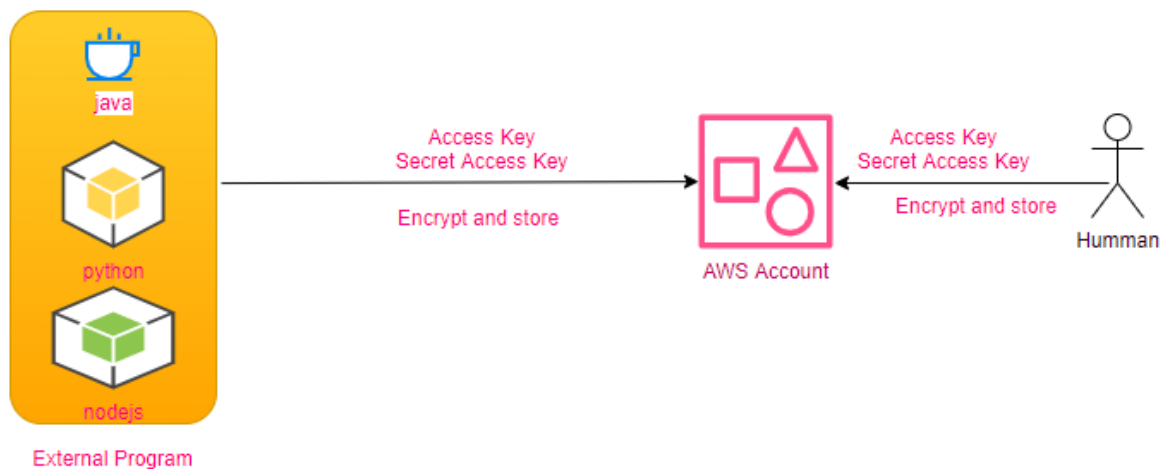
Select how these users will access AWS. Access keys and autogenerated passwords are provided in the last step. [Learn more](#)

Access type* ☐ **Programmatic access**
Enables an **access key ID** and **secret access key** for the AWS API, CLI, SDK, and other development tools.

☐ **AWS Management Console access**
Enables a **password** that allows users to sign-in to the AWS Management Console.

Access Modes:

- Programmatic(SDK,CLI): access key id, secret access key
- AWS Console: password



User Group: One than more Users

Policy: access permissions

“Statements”: { “Effect”: “Allow”,

“Action”: [“s3-outposts:Get*”, “s3-outposts:List*”],

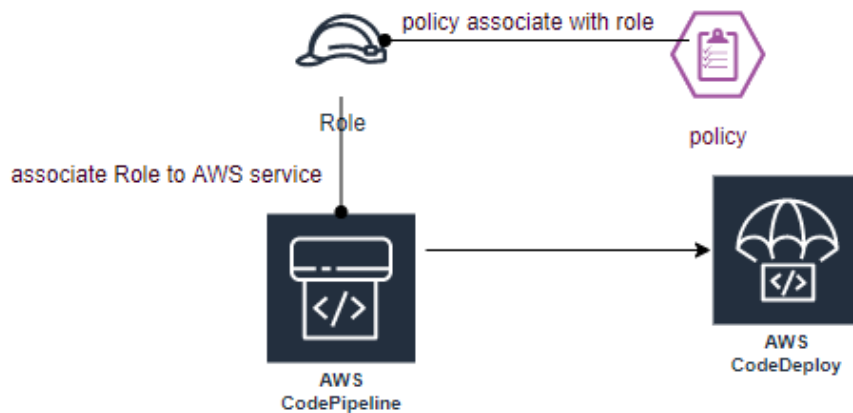
```

    "Resource": "*",
    "condition": {tag condition}
  },
] //statements

```

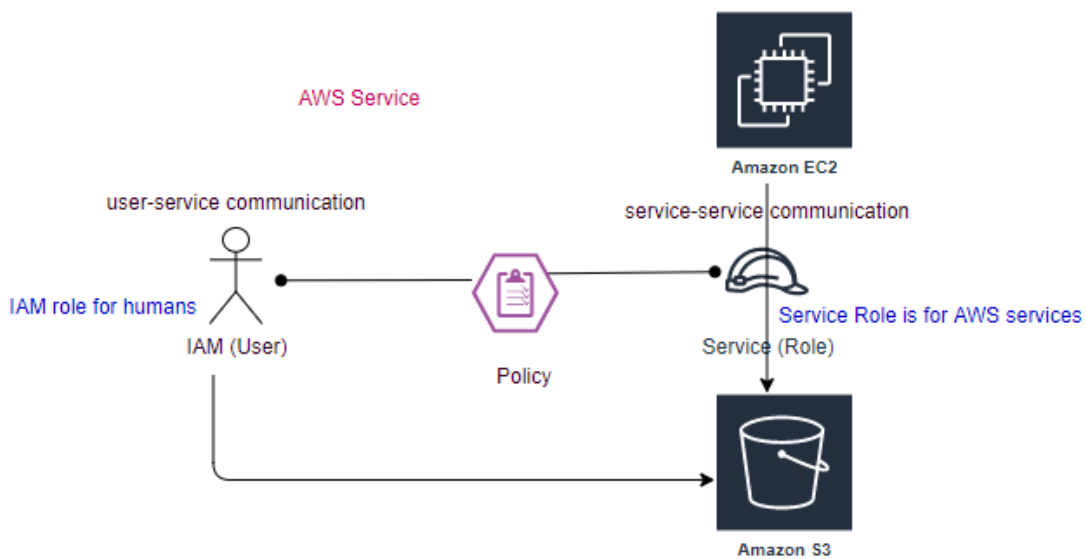
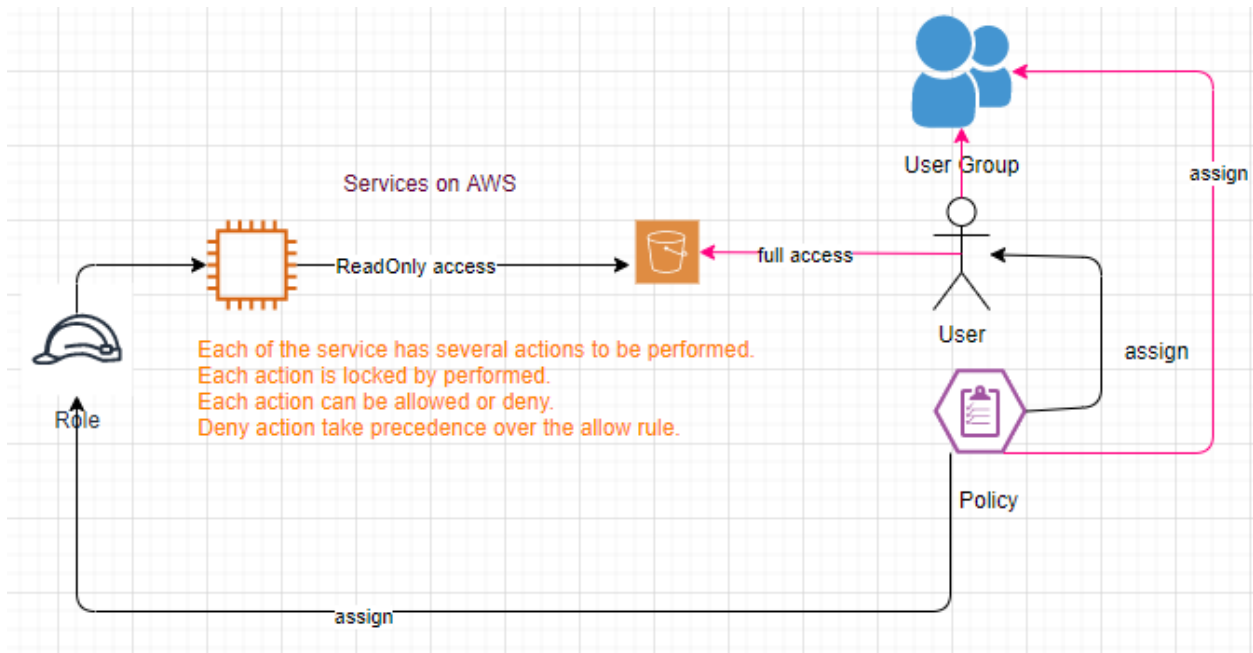
Roles: (Initiater of the Role)

Two services (AWS resources) with in AWS must be communicate to each other through the Role.



When human trying to access AWS resources using: console / Access and Secret Key, when AWS resource trying to access another AWS resource using: role

Policies can be assigned to user/user group **or to** Role.



access key and secret key stored in .aws\credentials its only for CLI and SDK access. Its not for console access, its for external access.

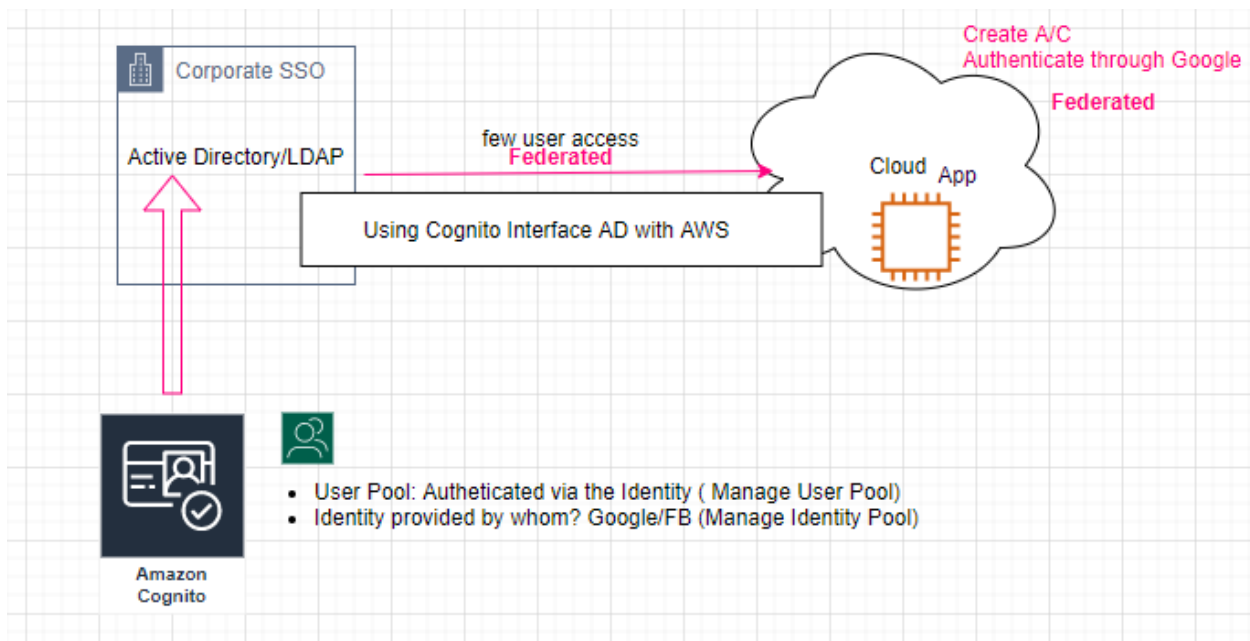
Policies are written in JSON (Java Script Object Notation).

Each service can provide different actions using policies can allow or deny specific action.

Billing Access

Tags and Usage -ABAC (Attribute Based Access Controls)

Federated?



Cognito-> Manage Identity Pool-> Create Identity Pool

▼ Authentication providers ⓘ

Amazon Cognito supports the following authentication methods with Amazon Cognito Sign-In or any public provider. If you allow your users to authenticate using any of these public providers, you can specify your application identifiers here. Warning: Changing the application ID that your identity pool is linked to will prevent existing users from authenticating using Amazon Cognito. [Learn more about public identity providers.](#)

Cognito

Amazon

Apple

Facebook

Google+

Twitter / Digits

OpenID

SAML

Custom

Amazon Cognito can authenticate users through any OpenID Connect provider. Once a provider has been configured with IAM, you can select the provider from the list below. [Learn more about using OpenId Connect providers.](#)

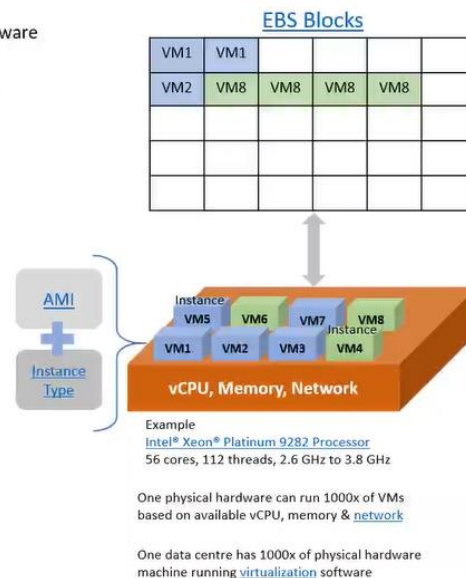
SAML for AD configuration.

EC2: (AMI): Amazon Machine Image

EC2 – Elastic Cloud Compute

edureka!

- **AMI**
 - Amazon Machine Image. Preconfigured templates of an OS + additional software
- **Instance Type**
 - Various configuration for of CPU, memory, storage and networking capacity
- **Instance**
 - VM configured for computing purpose using AMI + Instance Type
 - [Instance State](#)
- **EBS**
 - Elastic Block Storage, persistent storage volume for storing data
- **Instance Store**
 - Temporary data store, deleted when an instance is stopped or terminated
- **IPv4 address**
 - Private, Public and Elastic
- **[Security Group](#)**
 - Instance firewall, network defence layer along with subnet level NACL
- **[User Data](#)**
 - Set of commands executed on an EC2 instance when it is launched



AMI: O/S + additional s/w

- Root Device Type (boot drive): ebs (persistent)/ instance store(temporary)
- Virtualization Type: hvm/paravirtual
- ENA enabled: Yes/No

Instance Type: Various configuration for CPU, memory, storage and network capacity.

Instance Store: Temporary (transitory work load)

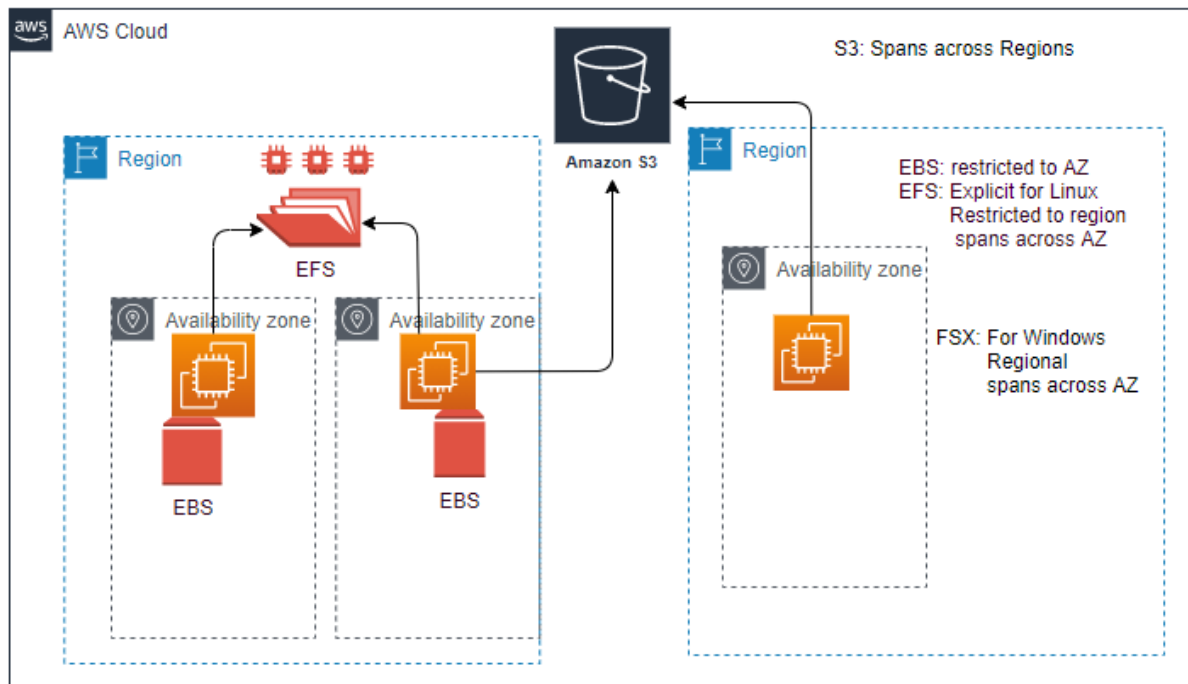
EBS: (Elastic Block Storage): Can Only Increase the Storage, but can't reduce, have boot drive.

Logical Unit Number (LUN)

Difference between EBS enabled (storage) **Instance Store** enabled devices

O/S has to load from some where, It is a temporary drive as soon as we stop the device instance type drive is revoked.

Storage Type:



Status Checks

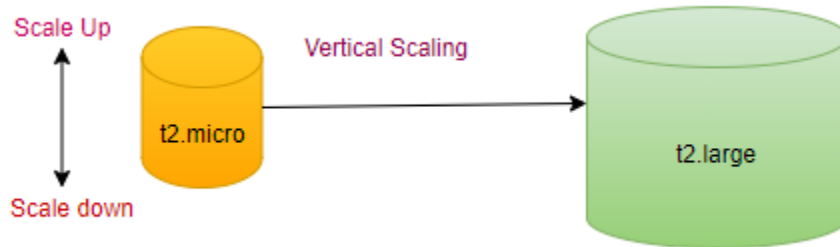
System Check: This check verifies that your instance is reachable. We test that we are able to get network packets to your instance. If this check fails, there may be an issue with the infrastructure hosting your instance (such as AWS power, networking or software system). You may need to restart or replace the instance, wait for our system to resolve the issue, or seek technical support. This check does not validate that your operating system and applications are accepting traffic. (Underneath configuration is available and not corrupted)

Instance Status Check: Once System check is passed, this check verifies that your instance's O/S is accepting traffic. If this check fails, you may need to reboot your instance or make modification to your operating system configuration. (Is reachable from outside the world)

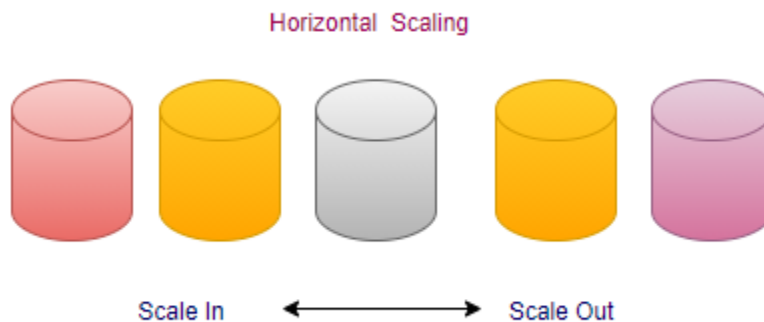
Scaling:

Vertical Saling: Stop (update from t2.micro->t2.small) , and then restart.

Can't vertical scale instance type at run time, first stop the instance then vertical scale the type.



Horizontal Saling: automate with zero downtime, Increase the number of resources of same capacity.



User Data:

Set of commands executed on the EC2 instance when it is launched (boot startp).

step1: choose the EC2 instance.

step1.1: right click-> Monitor and troubleshoot-> Get System log

All communication with in subnet through private IP.

HA through multi-AZ.